

Shellforge

Rollno: 2520030011

Program:

```
tejaswini@teju: ~/Skill
GNU nano 8.7.1 Shellforge.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <dirent.h>
#include <sys/stat.h>
#include <time.h>
#include <limits.h>
#include <errno.h>

void showHelp() {
    printf("\n");
    printf("=====\n");
    printf("          SHELLFORGE COMMANDS\n");
    printf("=====\n");
    printf(" 1. ls          List files and directories\n");
    printf(" 2. pwd         Show current directory\n");
    printf(" 3. mkdir <name> Create a new directory\n");
    printf(" 4. cd <name>    Change directory\n");
    printf(" 5. date        Show current date and time\n");
    printf(" 6. cal         Show calendar\n");
    printf(" 7. touch <file> Create an empty file\n");
    printf(" 8. rm <file>    Delete a file\n");
    printf(" 9. clear       Clear terminal\n");
    printf("10. exit        Exit ShellForge\n");
    printf("=====\n");
}

int main() {
    char command[200];
    char hostname[100];
    char currentPath[PATH_MAX];

    char username[] = "2520030011_Tejaswini";

    if (gethostname(hostname, sizeof(hostname)) != 0) {
        strcpy(hostname, "localhost");
    }
}
```

```
}

printf("\n");
printf("=====\n");
printf("          WELCOME TO SHELLFORGE\n");
printf("=====\n");
printf("          A Simple Linux Command Shell\n");
printf("=====\n");
printf("Type 'help' to see available commands.\n");
printf("=====\n");

while (1) {

    if (getcwd(currentPath, sizeof(currentPath)) == NULL) {
        strcpy(currentPath, "?");
    }

    printf("\n");
    printf("\033[1;36m%s@\033[0m:\033[1;32m%s\033[0m$ ",
        username,
        hostname,
        currentPath);

    if (fgets(command, sizeof(command), stdin) == NULL) {
        break;
    }

    command[strcspn(command, "\n")] = '\0';

    if (strcmp(command, "ls") == 0) {

        DIR *dir;
        struct dirent *entry;

        dir = opendir(".");

        if (dir == NULL) {
            printf("Unable to open directory.\n");
            continue;
        }

        printf("\n");
        printf("Contents of current directory:\n");
        printf("-----\n");

        while ((entry = readdir(dir)) != NULL) {

            if (entry->d_name[0] != '.') {
                printf("%-25s", entry->d_name);
            }
        }

        printf("\n");
        closedir(dir);
    }
}
```

```

else if (strcmp(command, "pwd") == 0) {

    if (getcwd(currentPath, sizeof(currentPath)) != NULL) {
        printf("\nCurrent Directory:\n%s\n", currentPath);
    }
    else {
        printf("Unable to get current directory.\n");
    }
}

else if (strncmp(command, "mkdir ", 6) == 0) {

    char *dirname = command + 6;

    if (strlen(dirname) == 0) {
        printf("mkdir: missing directory name\n");
    }

    else if (mkdir(dirname, 0777) == 0) {
        printf("Directory '%s' created successfully.\n", dirname);
    }

    else {
        if (errno == EEXIST) {
            printf("mkdir: '%s' already exists.\n", dirname);
        }
        else {
            printf("mkdir: unable to create directory '%s'.\n",
                dirname);
        }
    }
}

else if (strncmp(command, "cd ", 3) == 0) {

    char *dirname = command + 3;

    if (strlen(dirname) == 0) {
        printf("cd: missing directory name\n");
    }

    else if (chdir(dirname) == 0) {

        if (getcwd(currentPath, sizeof(currentPath)) != NULL) {
            printf("Changed directory to:\n%s\n", currentPath);
        }

    }

    else {
        printf("cd: unable to access '%s'\n", dirname);
    }
}

else if (strcmp(command, "date") == 0) {

    time_t currentTime;

    time(&currentTime);

    printf("\nCurrent Date and Time:\n");
    printf("%s", ctime(&currentTime));
}

```

```

    else {
        FILE *file = fopen(filename, "a");

        if (file == NULL) {
            printf("touch: unable to create '%s'\n", filename);
        }

        else {
            fclose(file);
            printf("File '%s' created successfully.\n", filename);
        }
    }
}

else if (strncmp(command, "rm ", 3) == 0) {
    char *filename = command + 3;

    if (strlen(filename) == 0) {
        printf("rm: missing file name\n");
    }

    else if (remove(filename) == 0) {
        printf("File '%s' deleted successfully.\n", filename);
    }

    else {
        printf("rm: unable to delete '%s'\n", filename);
    }
}

else if (strcmp(command, "clear") == 0) {
    system("clear");
}

else if (strcmp(command, "help") == 0) {
    showHelp();
}

else if (strcmp(command, "exit") == 0) {
    printf("\n=====\\n");
    printf("                Exiting ShellForge...\\n");
    printf("                Thank you for using ShellForge!\\n");
    printf("=====\\n");

    break;
}

else if (strlen(command) == 0) {
    continue;
}

else {
    printf("\nShellForge: command not found: %s\\n", command);
    printf("Type 'help' to see available commands.\\n");
}
}

return 0;
}

```

Output:

```
tejaswini@teju: ~/Skill
tejaswini@teju:~$ cd Skill
tejaswini@teju:~/Skill$ ls
tejaswini@teju:~/Skill$ nano Shellforge.c
tejaswini@teju:~/Skill$ gcc Shellforge.c -o Shellforge
tejaswini@teju:~/Skill$ ./Shellforge

=====
                        WELCOME TO SHELLFORGE
=====
                        A Simple Linux Command Shell
=====
Type 'help' to see available commands.
=====

2520030011_Tejaswini@teju:/home/tejaswini/Skill$ ls

Contents of current directory:
=====
Shellforge.c          Shellforge
=====

2520030011_Tejaswini@teju:/home/tejaswini/Skill$ help

=====
                        SHELLFORGE COMMANDS
=====
1. ls                  List files and directories
2. pwd                 Show current directory
3. mkdir <name>        Create a new directory
4. cd <name>           Change directory
5. date                Show current date and time
6. cal                 Show calendar
7. touch <file>        Create an empty file
8. rm <file>           Delete a file
9. clear               Clear terminal
10. exit                Exit ShellForge
=====

2520030011_Tejaswini@teju:/home/tejaswini/Skill$ pwd

Current Directory:
/home/tejaswini/Skill

2520030011_Tejaswini@teju:/home/tejaswini/Skill$ cal

    September 2026
Su Mo Tu We Th Fr Sa
                1  2  3  4  5
 6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30
```

```
2520030011_Tejaswini@teju:/home/tejaswini/Skill$ date
```

```
Current Date and Time:  
Wed Sep 30 02:41:41 2026
```

```
2520030011_Tejaswini@teju:/home/tejaswini/Skill$ mkdir temp  
Directory 'temp' created successfully.
```

```
2520030011_Tejaswini@teju:/home/tejaswini/Skill$ cd temp  
Changed directory to:  
/home/tejaswini/Skill/temp
```

```
2520030011_Tejaswini@teju:/home/tejaswini/Skill/temp$ exit
```

```
=====  
Exiting ShellForge...  
Thank you for using ShellForge!  
=====
```